

Paarth Sharma

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SOFTWARE ENGINEER HIGHLIGHTS

Software Engineering graduate with a Bachelor's in Software Engineering (Hons) and experience building Python/SQL data pipelines, FastAPI and React applications, AWS/Docker cloud systems, and AI/ML solutions across credit risk modelling, meeting intelligence, and quantitative backtesting.

- Built and deployed full-stack applications using React, TypeScript, FastAPI, PostgreSQL, Supabase, Docker, GitHub Actions, Vercel and Render to deliver authenticated, responsive and production-style web applications.
- Developed an XGBoost-based credit risk model at Origin Energy using Python, SQL, feature engineering and SHAP analysis on 100k+ customer records, achieving approximately 80% precision on high-risk predictions
- Implemented AI-powered applications using FastAPI, Whisper/Gemini integrations and transcript-aware Q&A workflows to automate transcription, summarisation, action-item extraction and information retrieval from meeting audio.
- Designed cloud-native and data-driven systems using AWS, Docker, REST APIs, CI/CD workflows and relational database modelling to support scalable deployment, backend reliability and end-to-end product delivery.

SKILLS: Python | TypeScript | JavaScript | Java | C/C++ | SQL | FastAPI | React | Next.js | Node.js | PostgreSQL | MySQL | MongoDB | Supabase | AWS | Docker | GitHub Actions | REST APIs | SQLAlchemy | Pandas | NumPy | scikit-learn | XGBoost | SHAP | LangChain | Tailwind CSS | Git | Linux

EDUCATION

MONASH UNIVERSITY

Bachelor of Software Engineering (Honours)

- Graduated with Distinction.
- Sir John Monash International Excellence Scholarship recipient.
- Faculty of Engineering Dean's Honour List 2022.

MELBOURNE, VIC

July 2021 – July 2025

EXPERIENCE

MONASH UNIVERSITY

Engineering Lab Tutor

MELBOURNE, VIC

July 2023 – Present

- Delivered Python programming, Arduino and embedded systems labs for 100+ engineering students, using hands-on coding exercises, hardware debugging and guided lab support to build practical engineering skills.
- Taught debugging, software fundamentals and systems thinking across 700+ hours of lab delivery by walking students through code errors, circuit behaviour and implementation trade-offs.
- Supported student projects by helping diagnose software, hardware and integration issues, improving students' ability to complete practical engineering tasks independently.
- Explained complex programming and embedded systems concepts to students with varying technical backgrounds by breaking problems into smaller examples, diagrams and step-by-step debugging workflows.

ORIGIN ENERGY

Customer Insights Intern

MELBOURNE, VIC

July 2024 – December 2024

- Built and maintained Python and SQL data pipelines using structured data extraction, cleaning and transformation workflows to process 10k–50k+ customer records for forecasting, reporting and analytics.
- Developed and evaluated an XGBoost-based credit risk model on 100k+ customer records using feature engineering, train/test validation and classification metrics, achieving approximately 80% precision in identifying high-risk customers.
- Engineered and validated customer risk features by iterating with credit collections stakeholders, improving model relevance and aligning outputs with business decision-making needs.
- Applied SHAP and feature importance analysis to explain model predictions, identify key default-risk drivers and improve stakeholder trust in model outputs.
- Optimised machine learning workflows by parallelising model execution with Python multiprocessing, reducing runtime by approximately 30%.
- Communicated analytical findings through stakeholder discussions and model output reviews to support production use cases for credit collections prioritisation.

FOCUS BEAR

Contract App Developer

MELBOURNE, VIC

August 2024 - September 2024

- Completed a short-term paid software contract contributing to an existing productivity application within a two-week time-boxed engagement.
- Investigated a targeted feature issue in an established codebase by working through local setup, dependency errors and debugging blockers while communicating progress and issues clearly.
- Adapted to an unfamiliar codebase and development environment by troubleshooting setup issues and clarifying blockers, strengthening practical experience with onboarding and remote technical collaboration.

NOTABLE PROJECTS

AI MEETING COPILOT

Personal Project

MELBOURNE, VIC

April 2026-May 2026

- Built a full-stack AI meeting assistant using Next.js, Tailwind, FastAPI and Python to convert uploaded meeting audio into structured transcripts, summaries, decisions and action items.
- Designed a backend processing workflow with FastAPI endpoints for audio upload, transcription handling and AI response generation, separating user-facing upload flows from server-side AI processing.
- Integrated Whisper-based transcription and Gemini-powered summarisation to reduce manual note-taking and make meeting information easier to review, search and reuse.

QUANT DEV BACKTESTER

Personal Project

MELBOURNE, VIC

March 2026 – April 2026

- Built a modular Python backtesting engine using Pandas, NumPy and yFinance to evaluate trading strategies against historical market data in a repeatable research workflow.
- Implemented reusable data ingestion, SMA feature engineering, signal generation and portfolio simulation modules to automate strategy testing and reduce manual analysis.
- Calculated Sharpe ratio, volatility, drawdown and strategy returns to compare strategies on risk-adjusted performance rather than raw returns alone.

MONASH UNIVERSITY

BirdTag - Cloud AI Project

MELBOURNE, VIC

April 2025 - May 2025

- Built a cloud-native AI application using AWS, Docker and machine learning models to detect bird species from uploaded image and audio recordings.
- Integrated YOLO object detection and BirdNET Analyzer into the application workflow, enabling multimodal species recognition across visual and audio inputs.
- Designed a distributed AWS architecture using Lambda, ECS containers, EC2, S3 and IAM to separate storage, compute and access control responsibilities.

MONASH UNIVERSITY

The Universal Book

MELBOURNE, VIC

Feb 2024 - May 2025

- Developed a full-stack collaborative publishing platform using React, Node.js and PostgreSQL to support digital book creation and multilingual content workflows by integrating Hugging Face translation models to automate translation.
- Designed PostgreSQL tables and relationships for users, book entries and translations, improving data consistency across collaborative publishing workflows.
- Containerised the application with Docker and deployed services on Nectar Cloud to improve environment consistency across development and deployment.

ORIGIN ENERGY

Collectability Risk Model

MELBOURNE, VIC

Nov 2024 - Dec 2024

- Developed an XGBoost classification model using Python, SQL and customer-level feature engineering to predict customers at risk of entering 90-day debt.
- Trained and evaluated the model on 100k+ customer records using classification metrics, achieving approximately 80% precision on high-risk predictions.
- Engineered domain-specific risk features through iterative collaboration with credit collections stakeholders, improving model relevance and alignment with business requirements.
- Applied SHAP and feature importance analysis to interpret model predictions, identify key risk drivers and support explainable decision-making for collections prioritisation.